User Journeys, Personas & User Stories

Human-Centred Safety UX & Adversary Evasion Workflows for South African GBV Survivors • Act 116/1998 & Act 14/2021 Compliant

1. Executive Context & Human-Centred Safety UX

In South Africa, the femicide rate is over five times the global average. Standard personal safety apps create severe dangers: abusers routinely inspect partners' phones, and discovering an emergency or SOS app frequently triggers retaliatory violence. Umkhuseli operates on the paradigm of **'Safety without suspicion; evidence without leakage'**. It presents an innocent daily utility layer (African proverbs launcher on mobile, municipal prepaid electricity calculator on web) that completely conceals a military-grade AES-256-GCM encrypted evidence vault and zero-latency emergency SOS beacon.

PHYSICAL THREAT MODEL

Core Threat Reality: The primary adversary is not a remote cybercriminal, but a hostile intimate partner in physical possession of the unlocked device. Every second of UI persistence or notification artifact can be life-threatening.

2. Target Personas & Stakeholder Profiles

Persona	Role & Demographics	Primary Needs & Constraints	Umkhuseli Solution Touchpoint
Naledi M.	28 yrs • Survivor of Intimate Partner Violence (IPV) • Soweto, Gauteng	Co-habits with abusive partner who audits her phone; requires zero visible safety traces and court-admissible diary.	Sankofa Proverbs Launcher facade; 0ms instant panic exit; AES-256-GCM evidence vault; Form 2 export.
Zanele K.	21 yrs • University Student • Durban commuter corridor, KZN	Travels high-risk public transit late at night; frequently has 0 data and 0 airtime balance.	Zero-airtime USSD distress pings (*120*7867#); offline 9-province Thuthuzela directory; 72-hr GPS beacon.
Lerato Khubeka	34 yrs • Primary ICE Contact (Sister / Guardian) • Johannesburg	Needs immediate, actionable SOS location alerts with handset telemetry to mobilize SAPS flying squad.	Receives automated SMS/WhatsApp alerts with Google Find My Device tracker and exact battery percentage.
Sister Dlamini	48 yrs • Forensic Nurse & Social Worker • Chris Hani Baragwanath TCC	Requires chronological, timestamped incident affidavits with SAPS OB references to assist court protection orders.	Receives standardized Form 2 Protection Order dossiers with Section 4(2) safe address withholding.

3. End-to-End User Story: 'Naledi's Shield'

A representative multi-stage operational journey demonstrating covert daily usage, evidence capture, distress escalation, and emergency evasion:

- **Phase 1: Normal Everyday Use (Innocent Cultural Facade)**

  Naledi uses her smartphone throughout the day. Umkhuseli functions as the 'Sankofa Wisdom Launcher', displaying inspiring Akan, Zulu, and Sotho proverbs with a full application drawer. When her partner inspects the device, no security or SOS app appears in the app drawer or running services.

- **Phase 2: Secret Entry & Biometric Evidence Logging**

When safe and in private, Naledi unlocks the Safe Haven via a covert gesture (triple-tapping the sacred Sankofa adinkra bird emblem or inputting secret PIN 1234/0000). Hardware BiometricPrompt verifies her fingerprint. Inside the Evidence Vault, she logs a physical altercation, attaches medical notes, and logs SAPS Occurrence Book OB 842/10/2026. Data is encrypted on-device via AES-256-GCM.

- **Phase 3: Immediate Threat Escalation & Tactile SOS Trigger**

A violent altercation begins in the household. Naledi triggers the Quick-Access SOS button. A 3-second tactile vibration countdown engages, allowing silent cancellation if triggered accidentally. Upon countdown completion, high-priority emergency SMS alerts with 72-hour live GPS tracks, battery telemetry, and Google Find My Device tracker URLs dispatch to her 5 ICE contacts.

- **Phase 4: Instant Panic Camouflage Wipe (0 ms)**

The perpetrator charges into the room demanding to see what she is doing. Naledi hits the hardware Back button. Safe Haven terminates instantly in **0 milliseconds**, scrubbing decrypted data from RAM and restoring the peaceful proverbs launcher. No screenshot or recents card remains in Android task overview.

- **Phase 5: Magistrates Court Protection Order Submission**

Accompanied by a TCC social worker, Naledi generates an official Form 2 Protection Order PDF directly from her vault. Her residential address is statutorily withheld under Section 4(2) of the Domestic Violence Act, protecting her from location leakage while presenting chronological, tamper-evident testimony.

4. Specialized Safety Scenarios

Scenario A: Coerced Unlock & Duress PIN (Code 9999)

If an abuser physically forces the survivor to enter her PIN, inputting the **Duress PIN (9999)** triggers a dual-action: 1) The app suppresses all Safe Haven UI and renders a convincing, fully functional utility decoy (generating an authentic Eskom/City Power STS token); 2) A silent high-priority distress beacon is dispatched in the background to the primary ICE guardian with live GPS telemetry.

Scenario B: Commuter Transit with Zero Airtime & Data

University student Zanele is stalked while walking to a taxi rank in Durban. She has 0 MB of mobile data and R0.00 airtime. Umkhuseli connects directly to South Africa's statutory zero-rated infrastructure: one-touch dialing for SAPS Flying Squad (10111), toll-free GBV Command Centre (0800 428 428), and zero-data USSD callback pings (*120*7867#) that request an emergency callback to her handset.

SOUTH AFRICAN LEGAL ADMISSIBILITY

Evidentiary Integrity Guarantee: Under the Domestic Violence Amendment Act 14 of 2021, electronic messages and digital records are fully recognized as admissible evidence in South African courts. Umkhuseli's immutable timestamps and OB linking prevent claims of fabrication.